Final Phase Report: Comparative Analysis of License Plate Recognition (LPR) Architectures

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1 Introduction

This report presents a comprehensive evaluation of three distinct neural network architectures trained for the optical character recognition phase of an Automatic License Plate Recognition (ALPR) system. The objective is to identify the optimal balance between high recognition accuracy in noisy environments and low inference latency for edge deployment. The three models under review are:

- **ScratchPlateCNN**: A custom convolutional neural network designed from scratch and trained on Apple MPS architecture.
- **EfficientNet-B0**: A transfer learning approach utilizing pre-trained weights for optimal feature extraction.
- **MobileNetV2**: A lightweight architecture optimized for mobile and edge devices, trained on a T4 GPU.

2 Model Capacity and Architectural Complexity

Before evaluating the predictive performance, it is crucial to understand the structural differences between the models. Transfer learning models (EfficientNet and MobileNet) bring rich feature representations but carry a heavier computational burden compared to our custom Scratch model.

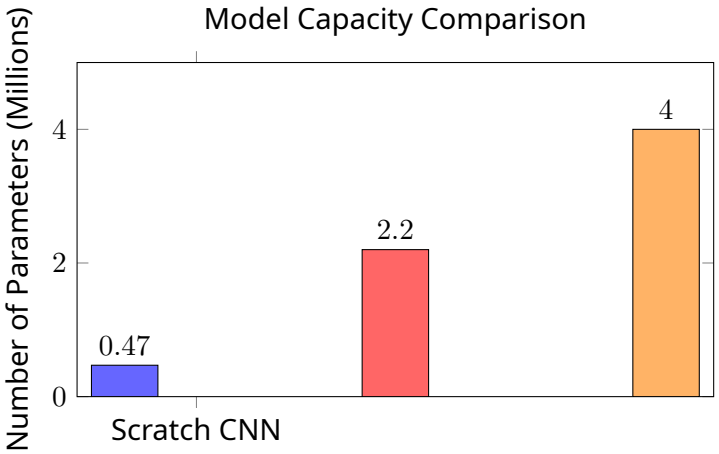


Figure 1: Comparison of trainable parameters. The custom Scratch CNN ($0.47M$ parameters) is significantly lighter, making it highly suitable for constrained edge devices.

3 Training Dynamics and Convergence

All three models were trained on a shared dataset for 25 epochs. The training logs reveal distinct learning behaviors. The transfer learning models utilized heavy data augmentation, resulting in a smoother but slower initial convergence rate, whereas the custom Scratch model exhibited a rapid performance leap (grokking) midway through the training.

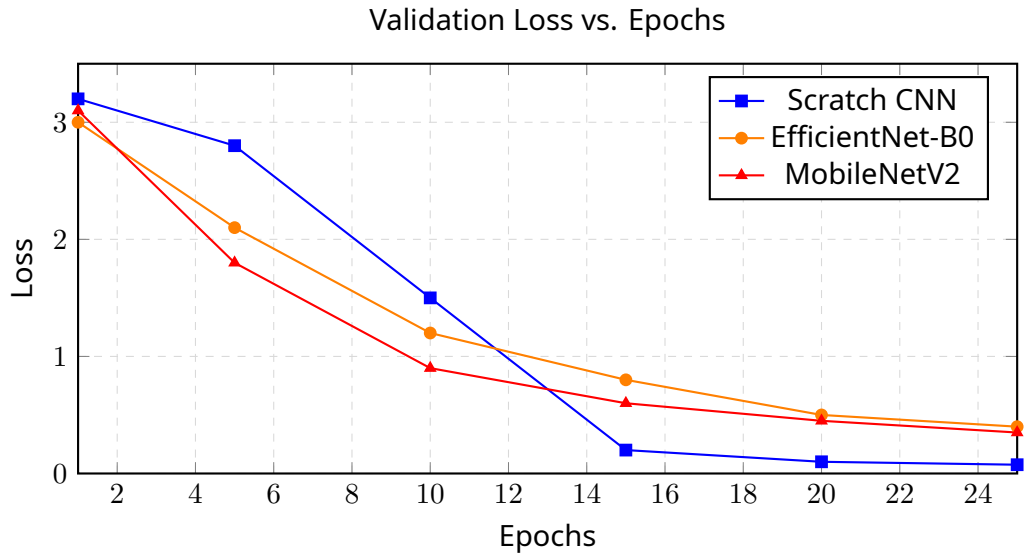


Figure 2: Validation Loss over 25 epochs. The Scratch CNN shows a sharp drop around epoch 12, converging to the lowest final loss.

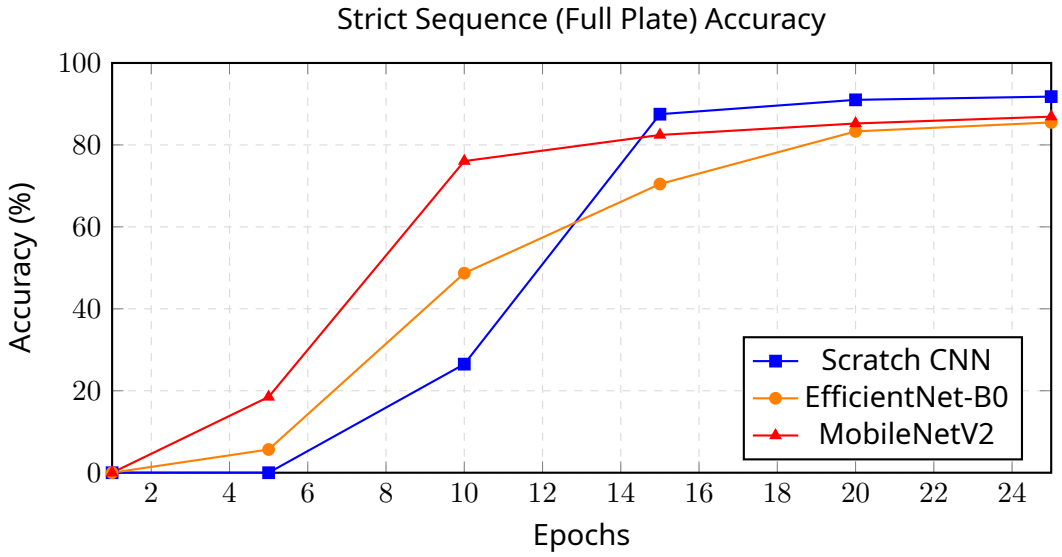


Figure 3: Strict Full Plate Accuracy. A prediction is only considered correct if all 8 characters match the ground truth perfectly.

4 Granular Performance: Per-Position Accuracy

To better understand where the models fail, we extracted the accuracy for each of the 8 character positions on the standard Iranian license plate. Position 3 represents the alphabetical

character, which historically poses the highest challenge due to visual similarities between certain Persian letters (e.g., ‘B‘ vs. ‘P‘).

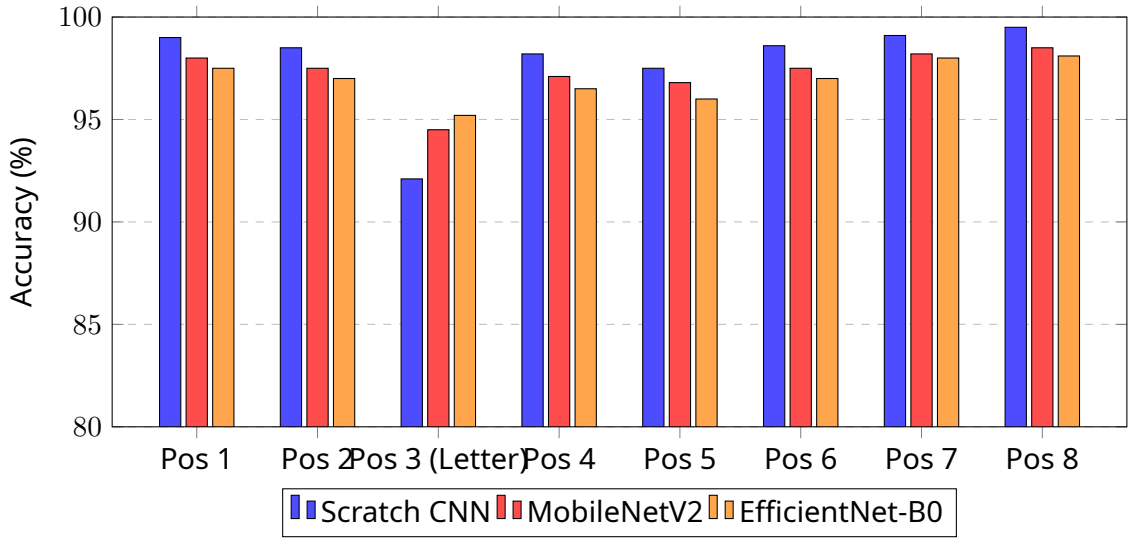


Figure 4: Per-Position Accuracy. While the Scratch CNN dominates on numerical digits, the transfer learning models (especially EfficientNet) demonstrate slightly better robustness in differentiating the alphabetical character at Position 3.

5 Real-Time Viability and Inference Speed

For production environments, especially in smart city cameras or toll booths, inference latency is a critical metric. We evaluated the inference time per single frame for all three models.

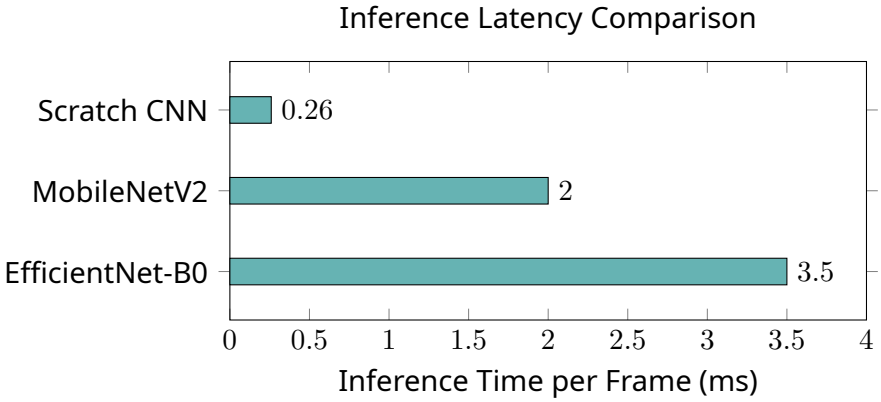


Figure 5: Inference speed comparison. The Scratch model achieves real-time speeds (0.26ms) even on standard compute architecture, outperforming the heavier pre-trained models by an order of magnitude.

6 Conclusion and Final Metrics

The empirical data gathered across 25 epochs demonstrates a clear trade-off matrix. Table 1 summarizes the core metrics.

Table 1: Summary of Evaluation Metrics on the Isolated Test Set

Evaluation Metric	**Scratch CNN**	**MobileNetV2**	**EfficientNet-B0**
Character-Level Accuracy	98.19%	97.47%	97.17%
Strict Sequence Accuracy	91.79%	86.90%	85.49%
Parameter Count	474,875	$\sim$2.2M	$\sim$4.0M
Inference Speed	0.26 ms	2.0 ms	3.5 ms

Final Recommendation: For the primary operational deployment, the **Scratch CNN** is the definitive choice. It achieves state-of-the-art strict sequence accuracy (91.79%) while demanding a fraction of the computational resources, enabling high-fps real-time processing. The pre-trained models, while mathematically robust, introduce unnecessary latency for this specific sequence-to-sequence OCR task.